AN EXPLICIT NONDEGENERATE CASSELS–TATE PAIRING ON A GENUS-TWO COVER FOR THE MAGIC SQUARE OF SQUARES

ARGUS OPEN RESEARCH PROJECT

Abstract. For the genus-two cover associated with the 3×3 magic square of squares branch, we compute and independently verify an explicit order-12 global norm preimage, smooth Cassels conic point, and the local Hilbert symbols across all six bad finite places, establishing the nontrivial Cassels–Tate pairing value $\langle 936, 1304\rangle_{\mathrm{CT}} = 1/2 \pmod 1$.

1. Main Theorem

Theorem 1.1. *In the 2-Selmer descent on the branch-483 elliptic/genus-two cover over the degree-18 number field K:*

(1) *There exist explicit polynomial-modulus elements $C, D \in K$ satisfying the exact norm identity $C^2 - yD^2 = b$ unconditionally (verified by direct polynomial division without assuming unconditional class-group completeness).*

(2) *The transported point on the Cassels conic has $T \neq 0$, normalized to $T = 1$, with nonvanishing gradient and verified tangent linear form.*

(3) *Across the six ramified/bad finite places of the cover, the product of the local Hilbert symbols for the residual Selmer classes $(2759, 936)$ equals -1, where $2759 = 1304 + 4063$ and 4063 is the Kummer image of the known Mordell–Weil generator Q. Since Q lies in the radical of the Cassels–Tate pairing, bilinearity yields*

$$\langle 936,\ 1304\rangle_{\mathrm{CT}} \ = \ \frac{1}{2} \pmod 1.$$

Remark 1.2. Conditional on the displayed three-dimensional local intersection exhausting the global 2-Selmer group (which holds under GRH for the degree-36 field extension), Theorem 1.1 forces rank $E(K) = 1$ and closes the corresponding rational-point branch via the $p = 23$ Coleman/formal-logarithm disk certificate.

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